1.2.2 Bhutasankhya Number System

In ancient India, there was a practice of using letters and words instead of numbers for calculations, counting, and representing days, etc. Among these Bhutasankhya and Katapayadi stand out as prominent number systems.

The *Bhūtasankhyā* system is one of the oldest ways of representing numbers in India. It is mentioned in early texts such as Pingala's *Chhanda Sūtra*. In this system, numbers are represented using meaningful words related to objects, natural elements, or ideas. These words are either directly linked to the number (like two eyes for 2) or symbolically connected through tradition or philosophy.

This method was widely used in oral traditions like poetry, astrology, and astronomy. Instead of writing numbers with digits, scholars used verses filled with these words. This helped people memorize and recite even long numbers easily, since meaningful words are easier to remember than digits.

Below are examples of some common number-word associations used in the Bhūtasakhyā system (see Data and Singh [3]):

- **0:** śūnya, kha, gagana, ambara, ākāśa, pūrṇa, ananta
- **1:** ādi, indu, candra, soma, bhūmi, ksiti, prṭhvī, mahī, rūpa
- **2:** yama, netra, drṣṭi, daśra, bāhu, kara, karṇa, osṭha, āyana
- **3:** guṇa, triguṇa, loka, bhuvana, kāla, trinetra, agni, purā
- **4:** veda, srṣṭi, samudra, sāgara, gati, diśā, varṇa, āśrama, yuga, traya āya, kosṭha
- **5:** śāstra, bāṇa, śara, bhūta, prāṇa, parva, tattva, indriya, ratna, vrata
- **6:** rasa, agni, kāya, rṭu, rāga, ari, śāstra, dravya

- **7:** nāga, aga, parvata, adri, giri, rṣi, yati, atri, vara, svara, dhātu
- **8:** vasu, ahi, nāga, gaja, danti, siddhi, dvīpa, sarpa, bhūti
- **9:** graha, anka, nanda, keśava, durgā, labdha, randhra
- **10:** diś, diśā, āśā, anguli, pañkti, avatāra, karma
- **11:** rudra, īśvara, īśa, bhava, bharga, mahādeva
- **12:** ravi, sūrya, ina, arka, martāṇḍa, bhānu, āditya, māsa, rāśi, vyaya
- **13:** viśva, kāma, atijagati
- **14:** manu, vidyā, indra, śakra, loka
- **15:** tithi, dina, abāna, paksa
- **16:** nrpa, bhūpa, bhupati, asti, kāla
- **27:** naksatra
- **32:** danta
- **33:** deva, amara
- **49:** tāna

The main idea of the Bhūtasankhyā system is straightforward: words that represent numbers are arranged in the same order as digits in the decimal system. However, these words are often written and read from right to left, instead of the usual left to right. In the next section, we will see examples of how such verses were used to express numbers symbolically and poetically.

We see below a few examples.

(i). Bhaskaracharya indicates the year of his birth and the date of composing the *Siddhant Siromani* through the following verse, making use of the bhuta-sankhya system.

रसगुणपूर्णमहीसमशकनृपसमये भवन्ममोत्पत्तिः । रसगुणवर्षेण मया सिद्धांतशिरोमणी रचितः ॥

रस = 6, गुण = 3, पूर्ण = 0, मही = 1

From the verse, Bhaskaracharya's birth year is given as शके 1036 (1114 CE). It also states that he wrote the *Siddhānta Śiromaṇi* at the age of 36(रसगु णवर्षे), that is, in 1150 CE.

(ii). Similarly, Aryabhata provides his birth year in Aryabhatiya as follows.

षष्ट्यब्दानां षष्टिर्यदा व्यतीतास्त्रयश्च युगपादाः । त्र्यधिकाविंशतिरब्दास्तदेदं मम जन्मनोऽतीतः ॥

षष्टि = 60,अब्द = year,षष्ट्यब्दानाम् = of 60 years,त्रयः = 3,विं शति = 20.

Aryabhata wrote that when 3600 years had passed and three-quarters of the yuga were over, he was 23 years old.

Historians generally agree that the Kaliyuga began in 3101 BCE. Using this reference, Aryabhata's birth year is calculated as

$$(3600 - 3101) - 23 = 476\,\text{CE}.$$

(iii). We will see in Chapter 4 a value of π obtained by Madhava correct upto 11 decimal places, expressed using bhutsankhya system.

1.2.3 Katapayadi Number System

The *Katapayadi* system is an old Indian method of writing numbers using letters. It helps in remembering large numbers by turning them into words or verses. This was useful in astronomy, mathematics, and poetry.

In this system, certain consonants are given number values from 0 to 9. These are arranged in four groups starting with the letters **ka, ta, pa, ya**. The name "Katapayadi" comes from the first letters of these groups. Vowels have no value and are used only to form pronounceable words. Below is a table showing how consonants map to digits:

1	2	3	4	5	6	7	8	9	0
क	ख	ग	घ	ङ	च	छ	ज	झ	ञ
ट	ठ	ड	ढ	ण	त	थ	द	ध	न
प	फ	ब	भ	म					
य	र	ल	व	श	ष	स	ह		Free Vow.

This system can be remembered using mnemonic phrases like "कादि नव तादि नव पादि पञ्चक् यादि अष्टक्". The Katapayadi system was mainly used in South India, especially in Kerala. It appears in the work of Vararuchi (4th century) in the text *Chandravakyam*, and later by Haridatta and Mādhava of Sangamagrama in astronomical texts like *Venuaroha*. The system helped encode numbers in poetic form for easy oral memorization.

This system can be remembered using mnemonic phrases like "कादि नव तादि नव पादि पञ्चक् यादि अष्टक्". The Katapayadi system was mainly used in South India, especially in Kerala. It appears in the work of Vararuchi (4th century) in the text *Chandravakyam*, and later by Haridatta and Mādhava of Sangamagrama in astronomical texts like *Venuaroha*. The system helped encode numbers in poetic form for easy oral memorization.

In the early version of this system, the number created from the letters was read from right to left. But in later versions, it was read from left to right, like we usually read numbers today.

The value of π to 17 decimal places: The value of π upto 17 decimal places is encoded using Katapayadi system in the following verse:

भद्राम्बुधिसिद्धजन्मगणितश्रद्धा स्म यद् भूपगीः

We decode it in the following table, where the first rows shows the original syllables, the second row shows only the relevant consonants, and the third row shows the digits.

भ	द्रा	म्बु	धि	सि	ध्द	ज	न्म	ग	णि	त	श्र	ध्दा	स्म	य	भ्दू	प	गीः
भ	र	ब	ध	स	ध्	ज	म	ग	ण	त	र	ध	म	य	भ्	प	ग
4	2	3	9	7	9	8	5	3	5	6	2	9	5	1	4	1	3

Readingthenumbersinthethirdrowinreverseordergivesthevalueof π as 3.14159265358979324.

Application to Carnatic classical music:

An interesting use of the Katapayadi system is found in Carnatic classical music (कर्नाटक शास्त्रीय संगीत), where it is applied to the names of rāgas(राग) . In this system, the first two syllables of a rāga name are assigned numerical values using the Katapayadi rules. When these digits are reversed, they indicate the Melakarta number of the rāga. The Melakarta system is a framework of 72 parent rāgas, each with a specific scale of notes.

For example, in the rāga "Mechakalyāni"(मे चकल्याणी), the first two syllables are "ma" and "cha", which correspond to the numbers 5 and 6. Reversing them gives 65, meaning Mechakalyāni is the 65th Melakarta rāga. This clever use of numbers connects the name of the rāga to its musical structure, showing how Indian knowledge traditions beautifully combine language, numbers, and music.

Madhava's value of π

Madhava (1340–1425) discovered an infinite series for π, which much later became known in Europe as the Gregory–Leibniz series. Using this series, Madhava was able to calculate π accurately up to 11 decimal places. Nilakantha Somayaji, writing in the 15th–16th centuries, records a verse composed by Madhava that encodes the value of π correct to 11 decimal places.

विबुधनेत्रगजाहिहुताशनत्रिगुणवेदभवारणबाहवः। नवनिखर्वमिते वृतिविस्तरे परिधिमानमिदं जगदुर्बुधाः॥

The first line of the verse gives the numbers in words: विबु ध (33); नेत्र (2); गज (8); अहि (8); हुताशन (3); त्रि (3); गुण (3); वेद (4); भवारण (27); बाहवः (28). Now, writing the numbers above in reverse order gives 2827433388233. This, as the verse explains, is the circumference of a circle whose diameter is equal to 9 nikharva, i.e. 9×10^{11}.
This implies that

$$\pi \approx \frac{2827433388233}{9 \times 10^{11}} = 3.1415926355822$$